AMERY ICE SHELF BOREHOLE MICROCAT DATA 2007/2008 SEASON - DATA PROCESSING AND QUALITY

Mark Rosenberg (data processor)
Amery Ice Shelf borehole drill team (data collectors)

INTRODUCTION

This report summarises the latest AMISOR microcat data, season 2007/2008, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes, including details of data processing and data quality. Temperature/salinity/pressure time series data were downloaded from three strings of Seabird SBE37IM microcats at AMISOR boreholes AM01, AM03 and AM04 (~13 months of new data from each site). Download from the microcats at borehole AM02 was attempted, however the microcat batteries were flat and data were unrecoverable.

All downloads were done by Mark Milnes, and a complete description of field operations is given in the report:

AMISOR Operations - 2007/2008, Mark Milnes, Australian Antarctic Division unpublished report,16pp.

Table 1: Station details for AMISOR season 2007/2008 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values over the data dates.

borehole	position at given date		microcat	mean instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM01	69°24.634'S	71°30.257'E	1969	436.2	441.1	07-Dec-2006	07-Jan-2008	30	+197
AM01	(on 07-Jan-2008)		1970	574.8	581.5	07-Dec-2006	07-Jan-2008	30	+184
AM01	"	"	1971	734.2	743.0	07-Dec-2006	09-Nov-2007	30	+209
AM02	69°41.198'S	72°43.182'E	1623	-	-	-	-	-	-
AM02	(on 23-Dec-2007)		1624	-	-	-	-	-	-
AM02	"	"	1174	-	-	-	-	-	-
AM03	70°33.271'S	70°20.163'E	4055	653.2	661.0	04-Dec-2006	04-Jan-2008	30	+190
AM03	(on 04-Jan-2008)		4054	861.0	871.7	04-Dec-2006	04-Jan-2008	30	+151
AM03	"	"	3883	1210.6	1226.6	04-Dec-2006	04-Jan-2008	30	+97
AM04	69°53.492'S	70°18.365'E	1972	542.1	548.4	25-Nov-2006	05-Jan-2008	30	+191
AM04	(on 05-Jan-2008)		1973	683.0	691.1	25-Nov-2006	05-Jan-2008	30	+232
AM04	"	"	1974	803.2	813.0	25-Nov-2006	05-Jan-2008	30	+123

MICROCAT DATA PROCESSING

The microcat moorings have already been described in previous reports. The mooring at AM01 was deployed in January 2002; January 2001 for AM02; December 2005 for AM03; and January 2006 for AM04. All latest microcat data downloaded during the 2007/2008 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time (Table 1). Clocks were reset to GPS time for all sets of microcats, following data download and prior to reinitialisation of logging.

Manufacturer supplied calibrations (May/June 2001 for AM01 instruments, June 2005 for AM03 instruments [except October 2004 for microcat-3883 pressure], and February 2003 for AM04 instruments) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* For AM01, pressure data were bad for all three instruments. A constant pressure value was selected for each AM01 instrument (Table 1), from inspection of data near the start of the deployment in January 2002. These pressure values were used to recalculate conductivity, and thence in the salinity calculations for each instrument.

* For AM03 and AM04, a mean pressure value was calculated for the year's data for each microcat (Table 1).

* Decimal times were “compressed” linearly throughout the time series to correct for clock error (Table 1).

MICROCAT DATA QUALITY NOTES

The following bad data were removed from the files:

* AM01, microcat 1970 - conductivity/salinity data spike removed for data points 18208, 19033 and 19034

* AM03, microcat 4055 - conductivity/salinity data spike removed for data point 12697

* AM03, microcat 4054 - conductivity/salinity data spike removed for data point 10494

* AM04, microcat 1972 - bad conductivity/salinity data removed for data points 10474-10543 and 14886-15175

The following data appear suspect (but not removed from the files):

* AM01, microcat 1969 - for first ~9 days, conductivity (and salinity) values slowly drift up, by ~0.02 mS/cm (Figure 1)

* AM01, microcat 1971 - conductivity and salinity values for the last patch of data from 5th to 9th November are suspect (too high)

* AM03, microcat 4055 - conductivity/salinity data are suspect for the following data points: 11912 to 11943; 16760 to 16762; 17403 to 17406

Microcat 1971 at AM01 was difficult to download, and only a small amount of data were recovered. This microcat appears to be on its last legs, with the batteries going flat.

Microcat data are plotted in Figures 1 to 4. The large conductivity and salinity data spikes evident in Figure 2 for microcat 4055 (AM03) are most likely due to fouling (ice crystals or critters).

For both AM03 and AM04, the pressure variation throughout the time series is very similar for all three microcats at each site (Figure 4). Although the order of the drift (~1 dbar at AM03 and ~2 dbar at AM04) is of the same order as the expected pressure sensor drift as quoted by the manufacturer, it is

unlikely that the pressure sensor calibration drift at each site would be so similar for each set of instruments - thus the pressure decrease at AM03 and AM04 over the year appears to be real. For AM01, unfortunately no good pressure data were available from the microcats to observe any pressure changes.

From conductivity and temperature sensor calibration drift estimates in a previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2006/2007 data downloads:

AM01 (in water for 6 years)	total S drift = 0.022 (PSS78)	total T drift = 0.0030^{o}C
AM03 (in water for 2 year)	total S drift = 0.007 (PSS78)	total T drift = 0.0010^{o}C

For AM04, manufacturer sensor calibrations were from February 2003, and the following total drifts might be expected at the time of the latest 2006/2007 data downloads (despite only 1 year in the water):

AM04 (after 5 years)	total S drift = 0.018 (PSS78)	total T drift = 0.0025^{o}C

Given the apparent real pressure changes observed at AM03 and AM04 (Figure 4), similar behaviour might be expected at AM01. So the lack of new pressure data from this site means there is an expected pressure error, which in turn would lead to a salinity error. In the worst case, this salinity error could potentially be up to 0.01 PSS, however without real pressure variation data at the site any estimate of this error is only a guess.

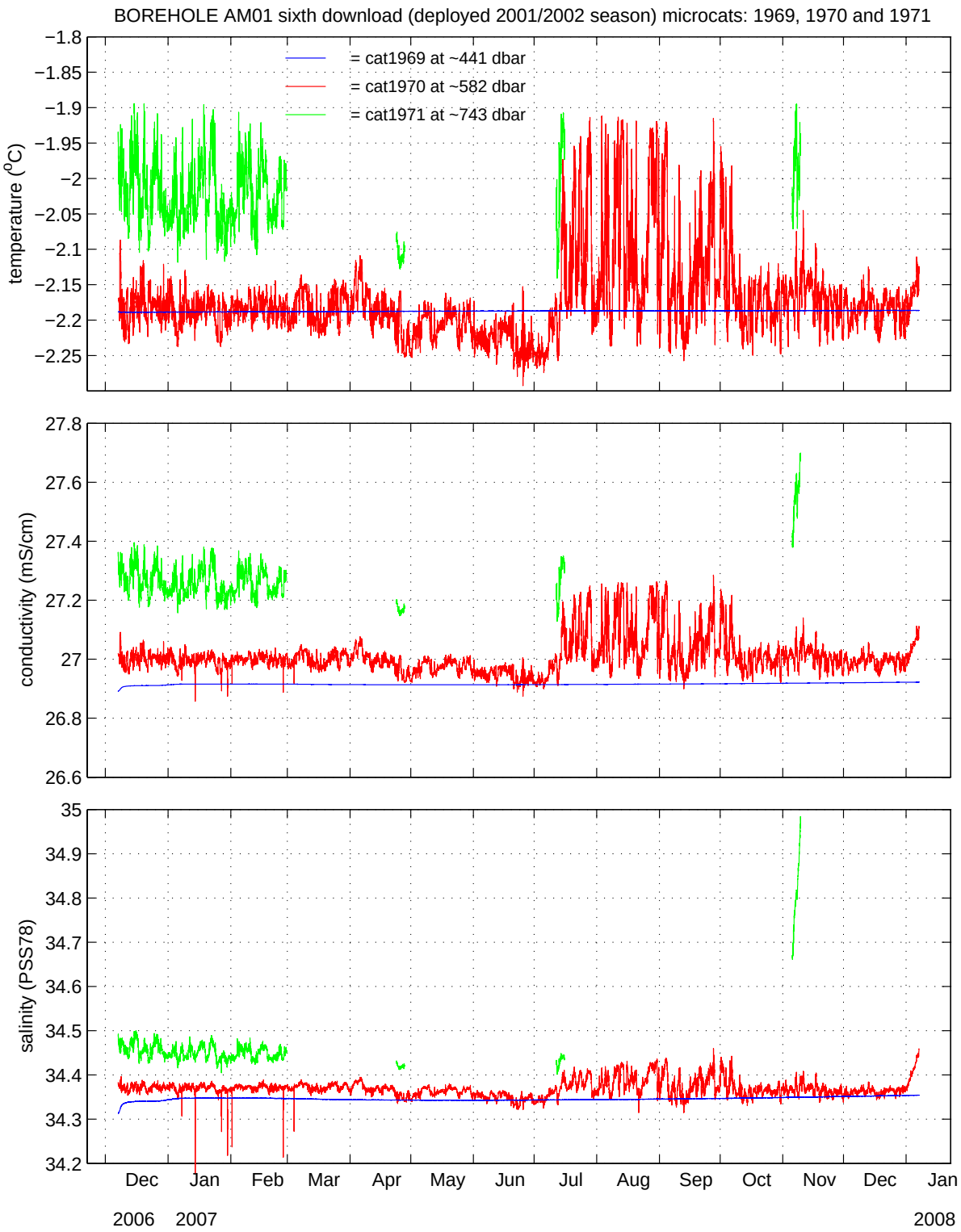


Figure 1: **Borehole AM01 microcat data downloaded in 2007/2008 season.**

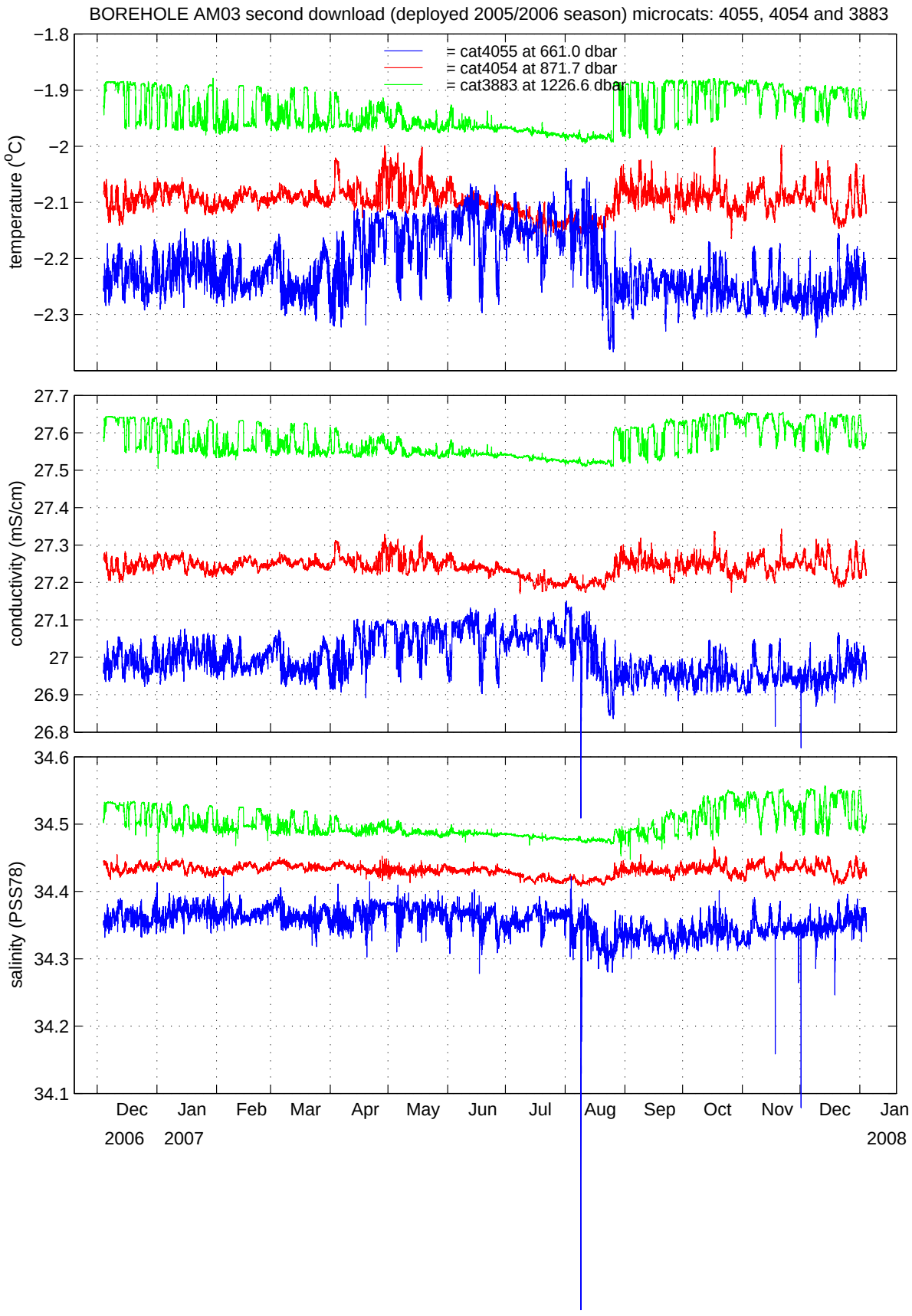
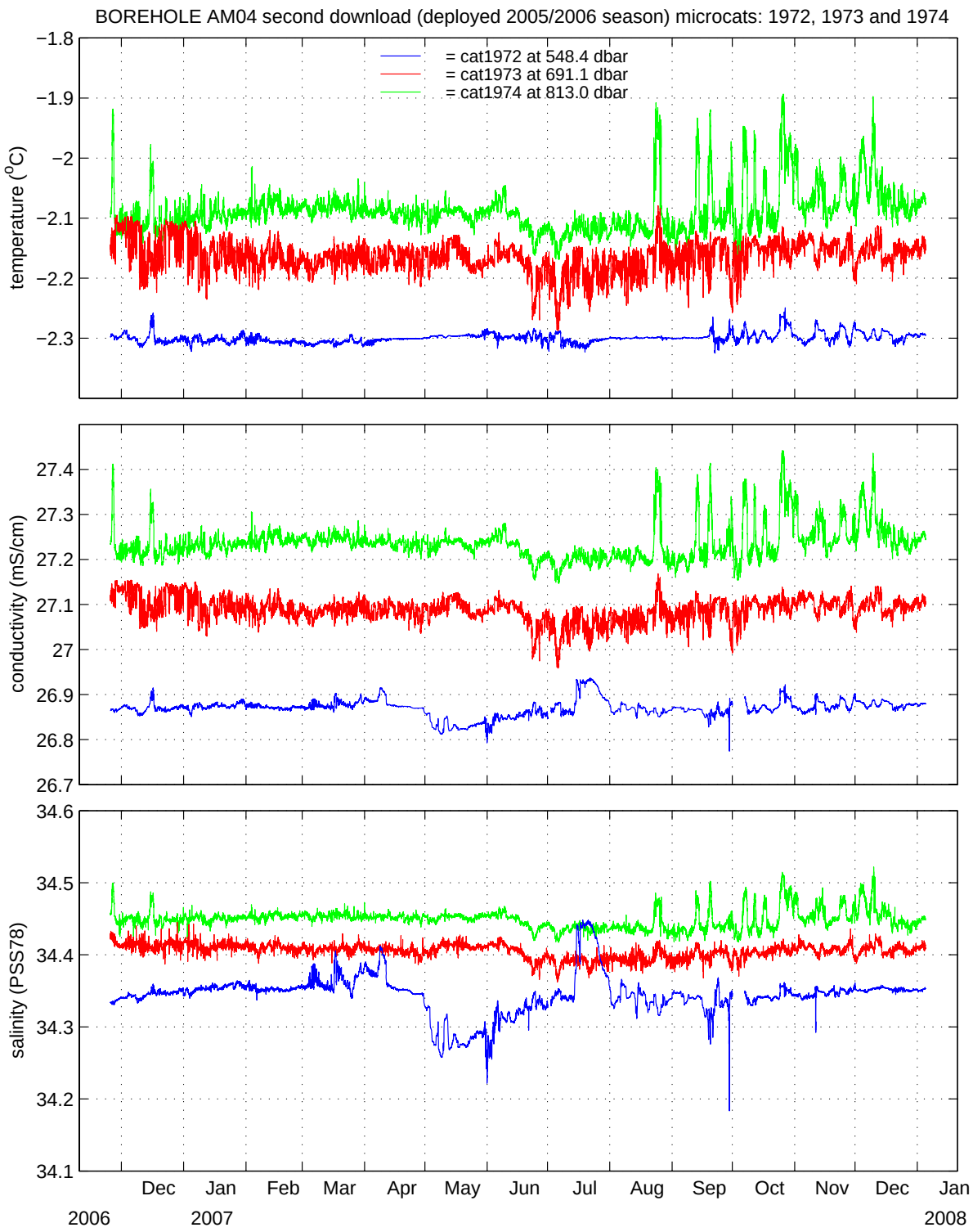
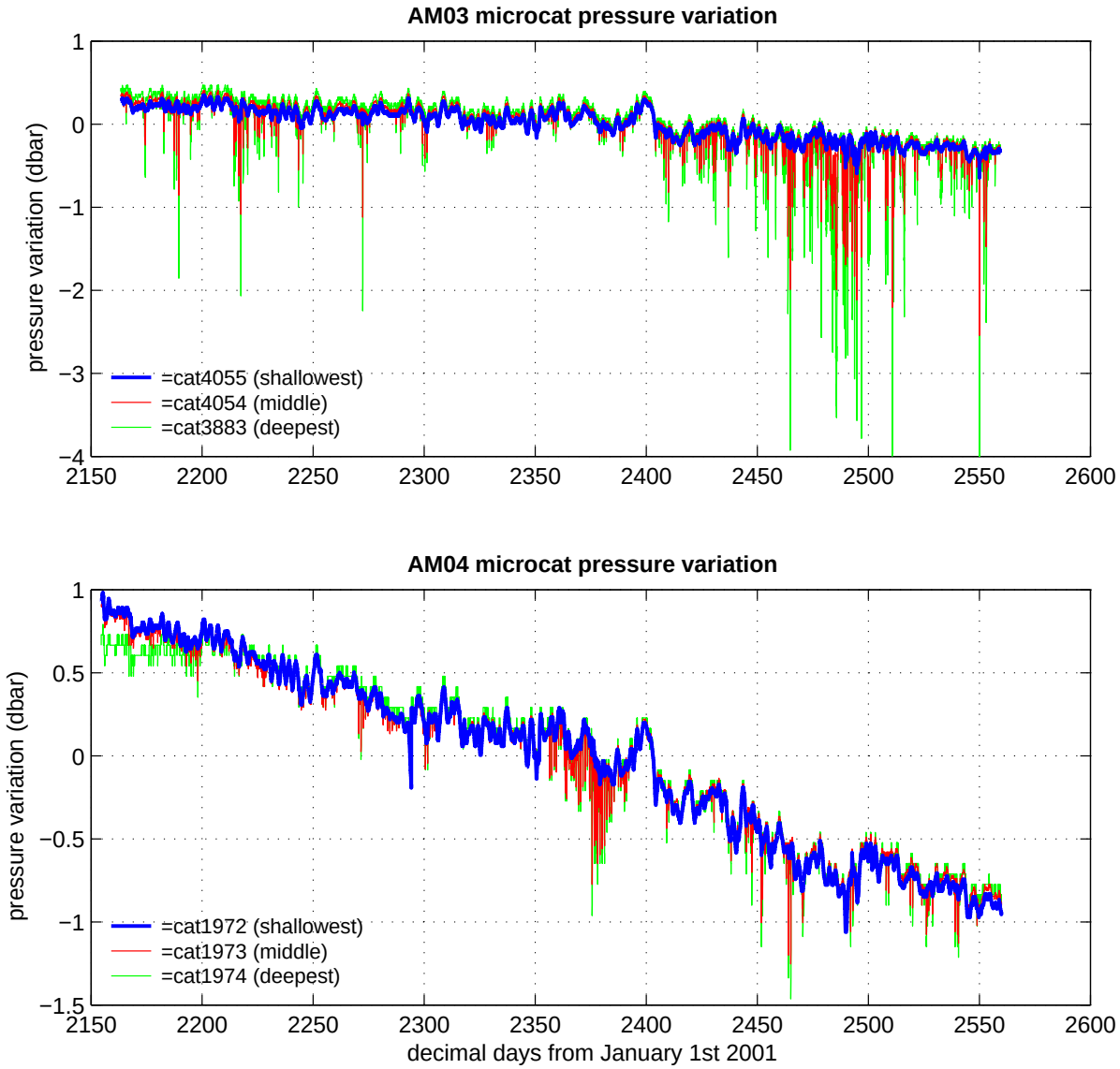


Figure 2: **Borehole AM03 microcat data downloaded in 2007/2008 season.**



<u>Figure 3:</u> Borehole AM04 microcat data downloaded in 2007/2008 season.



<u>Figure 4:</u> Pressure sensor data from boreholes AM03 and AM04 microcat data downloaded in 2007/2008 season. Plotted pressures are variations about the mean values.

DATA FILES

1. Microcat data from latest downloads

New microcat data from boreholes AM01, AM03 and AM04 are contained in text and matlab format files, as follows (note that there's no new AM02 data):

AM01
Data from the sixth data download (i.e. 07/12/2006-07/01/2008) are contained in the files:
cat1969_6.dat, cat1970_6.dat, cat1971_6.dat (text files), and
cat1969_6.mat, cat1970_6.mat, cat1971_6.mat (matlab files).
READMEcatam01_6 details these microcat file formats.

AM03
Data from the second data download (i.e. 04/12/2006-04/01/2008) are contained in the files:
cat4055_2.dat, cat4054_2.dat, cat3883_2.dat (text files), and
cat4055_2.mat, cat4054_2.mat, cat3883_2.mat (matlab files).
READMEcatam03_2 details the microcat file formats.

AM04
Data from the second data download (i.e. 25/11/2006-05/01/2008) are contained in the files:
cat1972_2.dat, cat1973_2.dat, cat1974_2.dat (text files), and
cat1972_2.mat, cat1973_2.mat, cat1974_2.mat (matlab files).
READMEcatam04_2 details the microcat file formats.

2. Unified microcat data files

The updated unified files (i.e. all data from initial deployment time, updated with data from the latest download) are contained in the following files:

AM01 (i.e. 15/01/2002-07/01/2008)
cat1969_all.dat, cat1970_all.dat, cat1971_all.dat (text files), and
cat1969_all.mat, cat1970_all.mat, cat1971_all.mat (matlab files).
READMEcatam01_all details these microcat file formats.
catamisor1all.eps is a time series plot.

AM02 (i.e. 06/01/2001-16/12/2006)
note: no new data here as all batteries are now flat
cat1623_all.dat, cat1624_all.dat, cat1174_all.dat (text files), and
cat1623_all.mat, cat1624_all.mat, cat1174_all.mat (matlab files).
READMEcatam02_all details these microcat file formats.
catamisor2all.eps and catamisor2allp.eps are time series plots.

AM03 (i.e. 27/12/2005-04/01/2008)
cat4055_all.dat, cat4054_all.dat, cat3883_all.dat (text files), and
cat4055_all.mat, cat4054_all.mat, cat3883_all.mat (matlab files).
READMEcatam03_all details these microcat file formats.
catamisor3all.eps and catamisor3allp.eps are time series plots.

AM04 (i.e. 11/01/2006-05/01/2008)
cat1972_all.dat, cat1973_all.dat, cat1974_all.dat (text files), and
cat1972_all.mat, cat1973_all.mat, cat1974_all.mat (matlab files).
READMEcatam04_all details these microcat file formats.
catamisor4all.eps and catamisor4allp.eps are time series plots.

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.